Curriculum

Aim

The programme aims to equip students with cutting-edge methodologies and skills in data analytics and artificial intelligence. By acquiring these competencies, students will be able to apply advanced technologies to transform business needs into real-world IT solutions.

Programme Characteristics

a. This programme integrates concepts and practices in the areas of data analytics and artificial intelligence to cope with the current and emerging market demands in the IT industry;
b. The programme offers optional overseas outreach learning activities (e.g., overseas internship, overseas workshop, overseas study tour, etc.), so that student could learn outside the classrooms to widen their horizons.

Programme Structure & Courses

The programme includes 14 units of core courses and 15 units of elective courses.

The elective courses are organized into: (i) Data Analytics, (ii) Artificial Intelligence, and (iii) MSc Practicum & Research. Students can select elective courses freely.

	unit(s)	unit(s)
Core Courses		**14**
Artificial Intelligence	3	
Python for Data Analytics and Artificial Intelligence	3	
Quantitative Methods for Data Analytics and Artificial Intelligence	3	
Principles and Practices of Data Analytics	3	
IT Forum	1	

Ethics of Data Analytics and Artificial Intelligence	1	
Elective Courses		**15**
Stream 1 – Data Analytics		
Innovative Laboratory	3	
Data Security and Privacy	3	
Blockchain and Cryptocurrenciesx	3	
Special Topics in Data Analytics	3	
Financial Technology	3	
Database Systems and Administration	3	
Data Mining and Knowledge Discovery	3	
Business Intelligence	3	
Visual Analytics and Decision Support	3	
Big Data Analytics	3	
Cloud Computing	3	
Audio and Speech Processing	3	
Stream 2 – Artificial Intelligence		
Natural Language Processing and Large Language Models	3	
Computer Vision	3	
Innovative Laboratory	3	
Prompt Engineering for Generative AI	3	
Blockchain and Cryptocurrencies	3	
Special Topics in Artificial Intelligence	3	
Recommender Systems	3	
Machine Learning	3	
Web Intelligence and Its Applications	3	
Data Mining and Knowledge Discovery	3	
Business Intelligence	3	
Audio and Speech Processing	3	
Autonomous Robotics	3	
MSc Practicum & Research		
MSc Practicum	3	
MSc Research I	3	
MSc Research II	3	
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Remarks: There may be minor revisions to the curriculum.

Duration & Schedule

The programme is offered in both one-year full-time mode and two-year part-time mode. Most classes are conducted on weekday nights at 6:30-9:20pm while a few classes, assessment activities and make-up classes may be conducted on Saturday daytime. In this manner, both full-time and part-time students can attend these classes/activities.

For full-time study, the normal study schedule is as follows:

	Semester 1 (units)	Semester 2 (units)
First Year		
Python for Data Analytics and Artificial Intelligence	3	

Principles and Practices of Data Analytics	3	
Artificial Intelligence	3	
Quantitative Methods for Data Analytics and Artificial Intelligence	3	
Ethics of Data Analytics and Artificial Intelligence	1	
IT Forum		1
Elective Courses		15

For part-time study, the normal study schedule is as follows:

	Semester 1 (units)	Semester 2 (units)
First Year		
Principles and Practices of Data Analytics	3	
Quantitative Methods for Data Analytics and Artificial Intelligence	3	
Ethics of Data Analytics and Artificial Intelligence	1	
Elective Courses		9
Second Year		
Python for Data Analytics and Artificial Intelligence	3	
Artificial Intelligence	3	
IT Forum		1
Elective Courses		6

Medium of Instruction

The medium of instruction is **English**. All teaching and learning materials are in English.

CONTACT US

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Note: The above information is correct to the best of our knowledge at the time of writing. In case of discrepancy, the information provided by the Hong Kong Baptist University and the Hong Kong SAR Government prevails.